Z - transform

Let $f(K)$ be a sequence.

One sided Z-transform,

$$Z\{f(K)\} = F(z) = \sum_{K=0}^{\infty} f(K)\, z^{-K}.$$

Two sided Z-tranform,

$$Z\{f(K)\} = \sum_{K=-\infty}^{\infty} f(k)\, z^{-K}.$$

[K is an integer i.e. $K = 0, \pm 1, \pm 2, \pm 3, \ldots$]

Q.1) Find Z-transform of the following sequence:

(i) $f(k) = \{15, 10, 7, 4, 1, -1, 0, 3, 6\}$
$f(-3)\ f(-2)\ f(-1)\ f(0)\ f(1)\ f(2)\ f(3)\ f(4)\ f(5)$

Soln:

$$Z\{f(k)\} = \sum_{k=-\infty}^{\infty} f(k)\, z^{-k}$$

$$= \sum_{k=-3}^{5} f(k)\, z^{-k}$$

$$= f(-3)z^3 + f(-2)z^2 + f(-1)z + f(0) + f(1)z^{-1} + f(2)z^{-2} + f(3)z^{-3} + f(4)z^{-4} + f(5)z^{-5}$$

$$= 15z^3 + 10z^2 + 7z + 4 + z^{-1} - 1z^{-2} + 0 + 3z^{-4} + 6z^{-5}$$

$$= 15z^3 + 10z^2 + 7z + 4 + \frac{1}{z} - \frac{1}{z^2} + \frac{3}{z^4} + \frac{6}{z^5}$$

(ii) $f(k) = \{5, 6, 1, 2, -1, 0, 8, 4, 3\}$

Soln:

$$Z\{f(k)\} = \sum_{k=0}^{\infty} f(k) z^{-k}$$

$$= f(0) + f(1) z^{-1} + f(2) z^{-2} + f(3) z^{-3} + f(4) z^{-4} + f(5) z^{-5} + f(6) z^{-6} + f(7) z^{-7} + f(8) z^{-8}$$

$$= 5 + \frac{6}{z} + \frac{1}{z^2} + \frac{2}{z^3} - \frac{1}{z^4} + \frac{8}{z^6} + \frac{4}{z^7} + \frac{3}{z^8}$$

(iii) $f(k) = \{1/4^k\}$

Soln:

$$Z\{f(k)\} = \sum_{k=-\infty}^{\infty} f(k) z^{-k}$$

$$= \sum_{k=-\infty}^{\infty} \frac{1}{4^k} z^{-k}$$

$$= \ldots \frac{1}{4^{-3}} z^3 + \frac{1}{4^{-2}} z^2 + \frac{1}{4^{-1}} z + \frac{1}{4^0} z^0 + \frac{1}{4^1} z^{-1} + \frac{1}{4^2} z^{-2} + \frac{1}{4^3} z^{-3} + \ldots$$

$$= 64 z^3 + 16 z^2 + 4z + 1 + \frac{1}{4z} + \frac{1}{16z^2} + \ldots$$

(iv) $f(k) = \left\{ \frac{1}{2^k} \right\}, \quad -3 \le k \le 3$

Soln:

$$Z\{f(k)\} = \sum_{k=-3}^{3} f(k)\, z^{-k}$$

$$= \sum_{k=-3}^{3} \frac{1}{2^k} \cdot z^{-k}$$

$$= f(-3)z^3 + f(-2)z^2 + f(-1)z^1 + f(0)z^0 + f(1)z^{-1} + f(2)z^{-2} + f(3)z^{-3}$$

$$= \frac{1}{2^{-3}} z^3 + \frac{1}{2^{-2}} z^2 + \frac{1}{2^{-1}} z^1 + \frac{1}{2^0} z^0 + \frac{1}{2^1} z^{-1} + \frac{1}{2^2} z^{-2} + \frac{1}{2^3} z^{-3}$$

$$= 8z^3 + 4z^2 + 2z + 1 + \frac{1}{2z} + \frac{1}{4z^2} + \frac{1}{8z^3}.$$

Some Important Z-transform Results:

$f(k)$, $k \geq 0$	$F(z) = Z\{f(k)\}$
$\delta(k)$	1
$U(k)$ or 1	$\frac{z}{z-1}$
k	$\frac{z}{(z-1)^2}$, $\lvert z \rvert > 1$
$k^n f(k)$	$\left(-z\frac{d}{dz}\right)^n F(z)$
k^n	$\left(-z\frac{d}{dz}\right)^n \cdot \frac{z}{z-1}$, $\lvert z \rvert > 1$
a^k or $a^k U(k)$	$\frac{z}{z-a}$, $\lvert z \rvert > \lvert a \rvert$
$\frac{1}{k}$	$\log\left(\frac{z}{z-1}\right)$, $\lvert z \rvert > 1$
$\frac{1}{k+1}$	$z \cdot \log\left(\frac{z}{z-1}\right)$
$\frac{1}{k!}$	$e^{1/z}$
$\delta(k-n)$	z^{-n}

$f(0)$ $\quad\quad$ $\lim_{z \to \infty} F(z)$

$\lim_{k \to \infty} f(k)$ $\quad\quad$ $\lim_{z \to 1} (z-1) F(z)$

$r_{f_1 f_2}(l) = \sum_{k=-\infty}^{\infty} f_1(k) . f_2(k-l)$ $\quad\quad$ $R_{f_1 f_2}(z) = F_1(z) . F_2(z^{-1})$

$h(k) = f(k) * g(k)$ $\quad\quad$ $F(z) . G_1(z)$

$f(-k)$ $\quad\quad$ $F(z^{-1})$

Unit Impulse Function

$$\delta(k) = \begin{cases} 1, & k = 0 \\ 0, & k \neq 0 \end{cases}$$

$$Z\{\delta(k)\} = \sum_{k=-\infty}^{\infty} f(k)\, z^{-k}$$

$$= \sum_{k=-\infty}^{\infty} \delta(k)\, z^{-k}$$

$$= \delta(0) : z^0$$

$$= 1.$$

Unit Step Function

$$U(k) = \begin{cases} 1, & k \geq 0 \\ 0, & k < 0 \end{cases}$$

$$Z\{U(k)\} = \sum_{k=-\infty}^{\infty} U(k) \cdot z^{-k}$$

$$= U(0) + U(1)z^{-1} + U(2)z^{-2} + U(3)z^{-3} + \ldots$$

$$= 1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \ldots$$

Here, G.P. infinite formula $= \frac{a}{1-r} = \frac{1}{1-\frac{1}{z}} = \frac{z}{z-1}$.

Change of Scale Property

If $Z\{f(k)\} = F(z)$ then

$Z(a^k f(k)) = F(z/a)$.

Find Z-transform of $a^k U(k)$

$$Z(a^k U(k)) = F(z/a) = \frac{z/a}{z/a - 1} = \frac{z}{z-a}$$

$$\boxed{Z\{U(k)\} = \frac{z}{z-1}}$$

Shifting Property

If $Z\{f(k)\} = F(z)$ then

$$Z\{f(k \pm n)\} = z^{\pm n} F(z)$$

Find Z-transform of $\delta(k-n)$

$$Z(\delta(k)) = 1 = F(z)$$

$$Z(\delta(k-n)) = z^{-n} F(z) = z^{-n}.$$

Division by k

If $Z\{f(k)\} = F(z)$ then

$$Z\left(\frac{f(k)}{k}\right) = -\int^{z} z^{-1} f(z)\, dz$$

Q.1) Find the Z-transform of $f * g$, $f(n) = U(n)$ & $g(n) = 2^n U(n)$ by convolution theorem.

Soln:

$$Z(f*g) = Z(f(n)) \cdot Z(g(n))$$

$$\Rightarrow Z(f*g) = Z(U(n)) \cdot Z(2^n U(n))$$

Now,

$$Z(U(n)) = \frac{z}{z-1}$$

$$Z(2^n U(n)) = \frac{z}{z-2}$$

$$Z(f*g) = \frac{z}{z-1} \cdot \frac{z}{z-2}$$

$$= \frac{z^2}{(z-1)(z-2)}$$

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

Q.2) Find Z-transform of $\sin(\alpha k)$, $k \geq 0$.

Soln:

$$Z(\sin(\alpha k)) = \sum_{k=0}^{\infty} \sin(\alpha k) \cdot z^{-k}$$

$$= \frac{1}{2i} \sum_{k=0}^{\infty} \left[e^{ik\alpha} - e^{-ik\alpha} \right] z^{-k}$$

$$= \frac{1}{2i} \left[\sum_{k=0}^{\infty} e^{ik\alpha} \cdot z^{-k} - \sum_{k=0}^{\infty} e^{-ik\alpha} \cdot z^{-k} \right]$$

$\left[\frac{e^{i\alpha}}{z} > 1 \ \& \ \frac{e^{ik}}{z} < 1 \right]$

$$= \frac{1}{2i} \left\{ \left[1 + \frac{e^{i\alpha}}{z} + \frac{e^{2iz}}{z^2} + \dots \right] - \left[1 + \frac{e^{-i\alpha}}{z} + \frac{e^{-2i\alpha}}{z} + \dots \right] \right\}$$

$$= \frac{1}{2i} \left\{ \frac{1}{\left(\frac{e^{i\alpha}}{z} - 1\right)} - \frac{1}{\left(1 - \frac{e^{-i\alpha}}{z}\right)} \right\}$$

$$= \frac{1}{2i} \left\{ \frac{z}{e^{i\alpha} - z} - \frac{z}{z - e^{-i\alpha}} \right\}$$

$$= \frac{z}{2i} \left[\frac{z - e^{-i\alpha} - e^{i\alpha} + z}{(e^{i\alpha} - z)(z - e^{-i\alpha})} \right]$$

$$= \frac{1}{2i}\left[\frac{1}{\left(1-\frac{e^{i\alpha}}{z}\right)} - \frac{1}{\left(1-\frac{e^{-i\alpha}}{z}\right)}\right]$$

$$= \frac{1}{2i}\left[\frac{z}{z-e^{i\alpha}} - \frac{z}{z-e^{-i\alpha}}\right]$$

$$= \frac{z}{2i}\left[\frac{z-e^{-i\alpha}+e^{i\alpha}-z}{(z-e^{i\alpha})(z-e^{-i\alpha})}\right]$$

$$= \frac{z}{2i}\left[\frac{e^{i\alpha}-e^{-i\alpha}}{z^2-ze^{-i\alpha}-ze^{i\alpha}+1}\right]$$

$$= \frac{z}{2i}\left[\frac{2\left(\frac{e^{i\alpha}-e^{-i\alpha}}{2}\right)}{z^2-2z\left(\frac{e^{i\alpha}+e^{-i\alpha}}{2}\right)+1}\right]$$

$$= \frac{z}{\cancel{2i}}\left[\frac{\cancel{2i}\sin\alpha}{z^2-2z\cos\alpha+1}\right]$$

$$\Rightarrow Z(\sin(\alpha k)) = \frac{z\sin\alpha}{z^2-2z\cos\alpha+1}.$$

Inverse Z-transform

If $Z\{f(k)\} = F(z)$ then

$$Z^{-1}\{F(z)\} = f(k).$$

Convolutional Method

If $Z(f * g) = Z(f(k)) . Z(g(k)) = F(z) . G(z)$

then

$$Z^{-1}(F(z) . G(z)) = f * g$$

$$= \sum_{m=0}^{k} f(m) . g(k-m).$$

Q. Find $Z^{-1}\left(\frac{z^2}{(z-a)(z-b)}\right)$.

Soln:

$$\frac{z^2}{(z-a)(z-b)} = \frac{z}{(z-a)} \cdot \frac{z}{(z-b)}$$

Here, $F(z) = \frac{z}{(z-a)}$ & $G(z) = \frac{z}{(z-b)}$

$$Z^{-1}\left(\frac{z^2}{(z-a)(z-b)}\right) = Z^{-1}(F(z).G(z))$$

$$= \sum_{m=0}^{k} f(m).g(k-m)$$

We have,

$$\left[Z(a^k) = \frac{z}{z-a}\right] \quad [f(k) = a^k \ \& \ g(k) = b^k]$$

$$Z^{-1}\left(\frac{z^2}{(z-a)(z-b)}\right) = \sum_{m=0}^{k} a^m b^{k-m}$$

$$= b^k + ab^{k-1} + a^2b^{k-2} + \ldots + a^k$$

$$= b^k\left[1 + \left(\frac{a}{b}\right) + \left(\frac{a}{b}\right)^2 + \ldots + \left(\frac{a}{b}\right)^k\right]$$

$$\left[\text{For G.P., } S_n = A.\frac{r^n-1}{r-1}\right]$$

$= b^k$ Here, $A=1$, $r = a/b$ & $n = k+1$

$$Z^{-1}\left(\frac{z^2}{(z-a)(z-b)}\right) = b^K \cdot 1\left[\frac{(a/b)^{K+1} - 1}{a/b - 1}\right]$$

$$= b^K \cdot \left(\frac{\frac{(a^{K+1} - b^{K+1})}{b^{K+1}}}{\frac{a-b}{b}}\right)$$

$$= \cancel{b^{K+1}} \cdot \left(\frac{a^{K+1} - b^{K+1}}{(a-b)\cancel{b^{K+1}}}\right)$$

$$= \frac{a^{K+1} - b^{K+1}}{a-b} .$$

Q.2) Find the inverse Z-transform of $F(z) = \frac{4z}{(z-a)}$

for (i) $|z| > |a|$ (ii) $|z| < |a|$

(i) Sol^n:

$$|z| > |a|$$

$$\Rightarrow \left|\frac{a}{z}\right| < 1$$

$$F(z) = \frac{4z}{z(1-a/z)} = 4\left(1-\frac{a}{z}\right)^{-1}$$

$$\Rightarrow F(z) = 4\left[1+\left(\frac{a}{z}\right)+\left(\frac{a}{z}\right)^2+\ldots.\right]$$

$$\Rightarrow F(z) = 4\sum_{n=0}^{\infty}\left(\frac{a}{z}\right)^n$$

$$\Rightarrow F(z) = 4\sum_{n=0}^{\infty} a^n . z^{-n}$$

$$\Rightarrow Z^{-1}(F(z)) = 4z^{-1}\left(\sum a^n z^{-n}\right)$$

$$\Rightarrow Z^{-1}(F(z)) = \{4a^k\}$$

$$Z(a^k) = \frac{z}{z-a}, \quad |z| > |a|$$

(ii) Soln:

$$|z| < |a|$$

$$\Rightarrow \left|\frac{z}{a}\right| < 1$$

$$F(z) = \frac{4z}{a\left(\frac{z}{a} - 1\right)} = -\frac{4z}{a}\left(1 - \frac{z}{a}\right)^{-1}$$

$$\Rightarrow F(z) = -\frac{4z}{a}\left(1 + \left(\frac{z}{a}\right) + \left(\frac{z}{a}\right)^2 + \ldots\right)$$

$$\Rightarrow (F(z)) = -4\left(\frac{z}{a} + \left(\frac{z}{a}\right)^2 + \left(\frac{z}{a}\right)^3 + \ldots\right)$$

$$\Rightarrow Z^{-1}(F(z)) = -4 \cdot Z^{-1}\left[\frac{z}{a} + \left(\frac{z}{a}\right)^2 + \left(\frac{z}{a}\right)^3 + \ldots\right]$$

$$\Rightarrow Z^{-1}(F(z)) = Z^{-1}\left(\frac{4z}{z-a}\right) = \{f(k)\}$$

Application of Z-transform - Difference Equation

y	x
y_1	x_1
y_2	x_2
y_3	x_3
y_4	x_4
⋮	⋮
y_n	x_n

$$\Delta y = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_2 - y_1}{h}$$

$$\Delta^2 y = \frac{y_3 - 2y_2 + y_1}{h^2}$$

$$\Delta^n y = \frac{y_{n+1} - 2y_n + y_{n-1}}{h^n}$$

$$Z(y_{K+n}) = z^n \left[\bar{y} - y_0 - \frac{y_1}{z} - \cdots \frac{y_{n-1}}{z^{n-1}} \right]$$

where, $Z(y_K) = \bar{y}$.

Q. Solve by z-transform:

$$y_{k+1} + y_k = 1 \quad \text{if} \quad y_0 = 0.$$

Soln:

$$Z(y_{k+1} + y_k) = Z(1)$$

$$\Rightarrow Z(y_{k+1}) + Z(y_k) = Z(1)$$

~~$\Rightarrow z^1[Z(y_k) - y_0$~~

$$\Rightarrow z^1[Z(y_k) - y_0] + z^0[Z(y_k)] = \frac{z}{z-1}$$

$$\Rightarrow z[Z(y_k)] + [Z(y_k)] = \frac{z}{z-1} \qquad [y_0 = 0]$$

$$\Rightarrow Z(y_k)[z+1] = \frac{z}{z-1}$$

$$\Rightarrow Z(y_k) = \frac{z}{(z+1)(z-1)}$$

$$\Rightarrow y_k = Z^{-1}\left[\frac{z}{(z+1)(z-1)}\right]$$

$$\frac{z}{(z+1)(z-1)} = \frac{z}{(z-1)} \cdot \frac{1}{(z+1)} = F(z) \cdot G(z)$$

$$Z^{-1}\left(F(z).G(z)\right) = \sum_{m=0}^{k} f(m).g(k-m)$$

$$\Rightarrow Z^{-1}\left(\frac{z}{(z-1)(z+1)}\right) = \sum_{m=0}^{k} f(m).g(k-m)$$

$$f(m) = Z^{-1}\left(F(z)\right) = Z^{-1}\left(\frac{z}{z-1}\right) = 1$$

$$g(k) = Z^{-1}\left(G(z)\right) = Z^{-1}\left(\frac{1}{z+1}\right) = (-1)^{k-1}\,U(k-1)$$

$$\left[Z^{-1}\left(\frac{1}{z-a}\right) = a^{k-1}\,U(k-1)\right]$$

$$\Rightarrow g(k-m) = (-1)^{k-m-1}\,U(k-m-1)$$

$$Z^{-1}\left(\frac{z}{(z-1)(z+1)}\right) = \sum_{m=0}^{k} (-1)^{k-m-1}\,U(k-m-1)$$

$$= (-1)^{k-1}\,U(k-1) + (-1)^{k-2}\,U(k-2) + \ldots + \ldots - U(-1)$$

$$\left[S_n = \frac{a(1-r^n)}{(1-r)}\right]$$

$$a = (-1)^{k-1}\,U(k-1)\,,\quad r = -1 < 1\,,\quad n = k+1$$

$$Z^{-1}\left(\frac{z}{(z-1)(z+1)}\right) = \frac{(-1)^{k-1}\,U(k-1)\left[1-(-1)^{k+1}\right]}{\left[1-(-1)\right]}$$

$$\Rightarrow Z^{-1}\left(\frac{z}{(z-1)(z+1)}\right) = \frac{(-1)^{k-1}\,U(k-1) - (-1)^{2k}\,U(k-1)}{2}.$$

$$\text{i.e. } y_k = \frac{(-1)^{k-1}\,U(k-1) - (-1)^{2k}\,U(k-1)}{2}.$$